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ARTIFICIAL INTELLIGENCE APPARATUS FOR PREDICTING POWER CONSUMPTION, AND METHOD THEREFOR

Abstract

The present disclosure relates to an artificial intelligence device and method capable of predicting an amount of power consumption occurring in a home, can obtain power consumption data for pre-registered electronic devices, check whether there is a specific electronic device for which the power consumption data has not been obtained, and if there is a specific electronic device for which the power consumption data has not been obtained, obtain mixed power data consumed in a current home, classify the obtained mixed power data into a plurality of individual power data, extracts individual power data matching the specific electronic device, estimate power consumption data for the specific electronic device based on the extracted individual power data, and predict an amount of power consumption in a home based on power consumption data for all pre-registered electronic devices.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an artificial intelligence apparatus and method for predicting an amount of power consumption occurring in a home.

BACKGROUND ART

[0002] In general, an energy storage system (ESS) refers to an apparatus that improves power usage efficiency by storing energy produced from renewable energy sources such as solar and wind power in a storage device (for example, a battery) and then supplying electricity when needed.

[0003] For these energy storage systems (ESS), it is important to accurately predict the amount of power consumption occurring in a home to improve energy efficiency.

[0004] In this way, in order to accurately predict the amount of power consumption within a home, it is necessary to utilize not only external environmental data but also the products used and usage patterns of customers within the home.

DISCLOSURE

Technical Problem

[0005] An object of the present disclosure is to solve the above-mentioned problems and other problems.

[0006] An object of the present disclosure is to provide an artificial intelligence apparatus capable of accurately predicting an amount of power consumption in a home by estimating power consumption data of electronic devices whose usage patterns are unknown using a neural network model.

[0007] In addition, an object of the present disclosure is to provide an artificial intelligence apparatus capable of providing accurate energy consumption information for each electronic device to a user by predicting an amount of power consumption in a home based on energy information and usage pattern information for the electronic device.

[0008] In addition, an object of the present disclosure is to provide an artificial intelligence apparatus capable of providing users with accurate, specific, and highly reliable home energy consumption information by predicting an amount of power consumption in a home based on power consumption data of each electronic device, user schedule data, and external environment data using a neural network model.

Technical Solution

[0009] An artificial intelligence apparatus according to an embodiment of the present disclosure includes a memory storing power consumption data for electronic devices in a home; and a processor predicting power consumption in a home based on the power consumption data, in which the processor can obtain power consumption data for pre-registered electronic devices, check whether there is a specific electronic device for which the power consumption data has not been obtained, and if there is a specific electronic device for which the power consumption data has not been obtained, obtain mixed power data consumed in a current home, classifies the obtained mixed power data into a plurality of individual power data, extract individual power data matching the specific electronic device, estimate power consumption data for the specific electronic device based on the extracted individual power data, and predict an amount of power consumption in a home based on power consumption data for all pre-registered electronic devices.

[0010] A method for predicting power consumption of an artificial intelligence apparatus according to an embodiment of the present disclosure may include obtaining power consumption data for a pre-registered electronic device; checking whether there is a specific electronic device for which

the power consumption data has not been obtained; obtaining mixed power data consumed in a current home if there is a specific electronic device for which the power consumption data has not been obtained; classifying the obtained mixed power data into a plurality of individual power data and extracting individual power data matching the specific electronic device; estimating power consumption data for the specific electronic device based on the extracted individual power data; and predicting the amount of power consumption in a home based on power consumption data for all the pre-registered electronic devices.

Advantageous Effect

[0011] According to one embodiment of the present disclosure, an artificial intelligence apparatus can accurately predict an amount of power consumption in a home by estimating power consumption data of an electronic device whose usage pattern is unknown using a neural network model.

[0012] In addition, according to one embodiment of the present disclosure, the artificial intelligence apparatus can provide accurate energy consumption information for each electronic device to the user by predicting an amount of power consumption in a home based on energy information and usage pattern information for the electronic device.

[0013] In addition, according to one embodiment of the present disclosure, the artificial intelligence apparatus can provide accurate, specific, and highly reliable home energy consumption information to the user by predicting an amount of power consumption in the home based on power consumption data of each electronic device, user schedule data, and external environment data using a neural network model.

Description

DESCRIPTION OF DRAWINGS

[0014] FIG. 1 illustrates an artificial intelligence apparatus according to an embodiment of the present disclosure.

[0015] FIG. 2 illustrates an artificial intelligence server according to an embodiment of the present disclosure.

[0016] FIG. 3 illustrates an artificial intelligence apparatus applied to an energy storage system according to one embodiment of the present disclosure.

[0017] FIG. 4 is a view for explaining an electronic device registration process for predicting an amount of power consumption according to one embodiment of the present disclosure.

[0018] FIGS. 5 and 6 are views for explaining an energy source estimation process of a neural network model according to an embodiment of the present disclosure.

[0019] FIG. 7 is a view for explaining a process of predicting an amount of power consumption in a home using a neural network model according to an embodiment of the present disclosure.

[0020] FIG. 8 is a view for explaining information on prediction results of an amount of power consumption in a home according to one embodiment of the present disclosure.

[0021] FIG. 9 is a flowchart for explaining a method for predicting power consumption of an artificial intelligence apparatus according to an embodiment of the present disclosure.

BEST MODE

[0022] Hereinafter, embodiments of the present disclosure are described in more detail with reference to accompanying drawings and regardless of the drawings symbols, same or similar components are assigned with the same reference numerals and thus overlapping descriptions for those are omitted. The suffixes “module” and “unit” for components used in the description below are assigned or mixed in consideration of easiness in writing the specification and do not have distinctive meanings or roles by themselves. In the following description, detailed descriptions of well-known functions or constructions will be omitted since they would obscure the disclosure in

unnecessary detail. Additionally, the accompanying drawings are used to help easily understanding embodiments disclosed herein but the technical idea of the present disclosure is not limited thereto. It should be understood that all of variations, equivalents or substitutes contained in the concept and technical scope of the present disclosure are also included.

[0023] It will be understood that the terms “first” and “second” are used herein to describe various components but these components should not be limited by these terms. These terms are used only to distinguish one component from other components.

[0024] In this disclosure below, when one part (or element, device, etc.) is referred to as being ‘connected’ to another part (or element, device, etc.), it should be understood that the former can be ‘directly connected’ to the latter, or ‘electrically connected’ to the latter via an intervening part (or element, device, etc.). It will be further understood that when one component is referred to as being ‘directly connected’ or ‘directly’ linked to another component, it means that no intervening component is present.

[0025] Also, throughout this specification, a neural network and a network function may be used interchangeably. The neural network may be constituted by a set of interconnected computational units, which may be generally referred to as “nodes”. These “nodes” may also be referred to as “neurons”. The neural network is configured to include at least two or more nodes. Nodes (or neurons) constituting neural networks may be interconnected by one or more “links”.

Artificial Intelligence (AI)

[0026] Artificial intelligence refers to the field of studying artificial intelligence or methodology for making artificial intelligence, and machine learning refers to the field of defining various issues dealt with in the field of artificial intelligence and studying methodology for solving the various issues. Machine learning is defined as an algorithm that enhances the performance of a certain task through a steady experience with the certain task.

[0027] An artificial neural network (ANN) is a model used in machine learning and may mean a whole model of problem-solving ability which is composed of artificial neurons (nodes) that form a network by synaptic connections. The artificial neural network can be defined by a connection pattern between neurons in different layers, a learning process for updating model parameters, and an activation function for generating an output value.

[0028] The artificial neural network may include an input layer, an output layer, and optionally one or more hidden layers. Each layer includes one or more neurons, and the artificial neural network may include a synapse that links neurons to neurons. In the artificial neural network, each neuron may output the function value of the activation function for input signals, weights, and deflections input through the synapse.

[0029] Model parameters refer to parameters determined through learning and include a weight value of synaptic connection and deflection of neurons. A hyperparameter means a parameter to be set in the machine learning algorithm before learning, and includes a learning rate, a repetition number, a mini batch size, and an initialization function.

[0030] The purpose of the learning of the artificial neural network may be to determine the model parameters that minimize a loss function. The loss function may be used as an index to determine optimal model parameters in the learning process of the artificial neural network.

[0031] Machine learning may be classified into supervised learning, unsupervised learning, and reinforcement learning according to a learning method.

[0032] The supervised learning may refer to a method of training an artificial neural network in a state in which a label for learning data is given, and the label may mean the correct answer (or result value) that the artificial neural network must infer when the learning data is input to the artificial neural network. The unsupervised learning may refer to a method of training an artificial neural network in a state in which a label for learning data is not given. The reinforcement learning may refer to a learning method in which an agent defined in a certain environment learns to select a behavior or a behavior sequence that maximizes cumulative compensation in each state.

[0033] Machine learning, which is implemented as a deep neural network (DNN) including a plurality of hidden layers among artificial neural networks, is also referred to as deep learning, and the deep learning is part of machine learning. In the following, machine learning is used to mean deep learning.

[0034] FIG. 1 illustrates an AI device **100** according to an embodiment of the present disclosure.

[0035] The AI device (or an AI apparatus) **100** may be implemented by a stationary device or a mobile device, such as a TV, a projector, a mobile phone, a smartphone, a desktop computer, a notebook, a digital broadcasting terminal, a personal digital assistant (PDA), a portable multimedia player (PMP), a navigation device, a tablet PC, a wearable device, a set-top box (STB), a DMB receiver, a radio, a washing machine, a refrigerator, a desktop computer, a digital signage, a robot, a vehicle, and the like.

[0036] Referring to FIG. 1, the AI device **100** may include a communication unit **110**, an input unit **120**, a learning processor **130**, a sensing unit **140**, an output unit **150**, a memory **170**, and a processor **180**.

[0037] The communication unit **110** may transmit and receive data to and from external devices such as other AI devices **100a** to **100e** and the AI server **200** by using wire/wireless communication technology. For example, the communication unit **110** may transmit and receive sensor information, a user input, a learning model, and a control signal to and from external devices.

[0038] The communication technology used by the communication unit **110** includes GSM (Global System for Mobile communication), CDMA (Code Division Multi Access), LTE (Long Term Evolution), 5G, WLAN (Wireless LAN), Wi-Fi (Wireless-Fidelity), Bluetooth™, RFID (Radio Frequency Identification), Infrared Data Association (IrDA), ZigBee, NFC (Near Field Communication), and the like.

[0039] The input unit **120** may acquire various kinds of data.

[0040] In this case, the input unit **120** may include a camera for inputting a video signal, a microphone for receiving an audio signal, and a user input unit for receiving information from a user. The camera or the microphone may be treated as a sensor, and the signal obtained from the camera or the microphone may be referred to as sensing data or sensor information.

[0041] The input unit **120** may acquire a learning data for model learning and an input data to be used if an output is obtained by using learning model. The input unit **120** may acquire raw input data. In this case, the processor **180** or the learning processor **130** may extract an input feature by preprocessing the input data.

[0042] The learning processor **130** may learn a model composed of an artificial neural network by using learning data. The learned artificial neural network may be referred to as a learning model. The learning model may be used to an infer result value for new input data rather than learning data, and the inferred value may be used as a basis for determination to perform a certain operation.

[0043] At this time, the learning processor **130** may perform AI processing together with the learning processor **240** of the AI server **200** of FIG. 2.

[0044] At this time, the learning processor **130** may include a memory integrated or implemented in the AI device **100**. Alternatively, the learning processor **130** may be implemented by using the memory **170**, an external memory directly connected to the AI device **100**, or a memory held in an external device.

[0045] The sensing unit **140** may acquire at least one of internal information about the AI device **100**, ambient environment information about the AI device **100**, and user information by using various sensors.

[0046] Examples of the sensors included in the sensing unit **140** may include a proximity sensor, an illuminance sensor, an acceleration sensor, a magnetic sensor, a gyro sensor, an inertial sensor, an RGB sensor, an IR sensor, a fingerprint recognition sensor, an ultrasonic sensor, an optical sensor, a microphone, a lidar, and a radar.

[0047] The output unit **150** may generate an output related to a visual sense, an auditory sense, or a

haptic sense.

[0048] At this time, the output unit **150** may include a display unit for outputting time information, a speaker for outputting auditory information, and a haptic module for outputting haptic information.

[0049] The memory **170** may store data that supports various functions of the AI device **100**. For example, the memory **170** may store input data obtained by the input unit **120**, learning data, a learning model, a learning history, and the like.

[0050] The processor **180** may determine at least one executable operation of the AI device **100** based on information determined or generated by using a data analysis algorithm or a machine learning algorithm. The processor **180** may control the components of the AI device **100** to execute the determined operation.

[0051] To this end, the processor **180** may request, search, receive, or utilize data of the learning processor **130** or the memory **170**. The processor **180** may control the components of the AI device **100** to execute the predicted operation or the operation determined to be desirable among the at least one executable operation.

[0052] When the connection of an external device is required to perform the determined operation, the processor **180** may generate a control signal for controlling the external device and may transmit the generated control signal to the external device.

[0053] The processor **180** may acquire intention information for the user input and may determine the user's requirements based on the obtained intention information.

[0054] The processor **180** may acquire the intention information corresponding to the user input by using at least one of a speech to text (STT) engine for converting speech input into a text string or a natural language processing (NLP) engine for acquiring intention information of a natural language.

[0055] At least one of the STT engine or the NLP engine may be configured as an artificial neural network, at least part of which is learned according to the machine learning algorithm. At least one of the STT engine or the NLP engine may be learned by the learning processor **130**, may be learned by the learning processor **240** of the AI server **200**, or may be learned by their distributed processing.

[0056] The processor **180** may collect history information including the operation contents of the AI apparatus **100** or the user's feedback on the operation and may store the collected history information in the memory **170** or the learning processor **130** or transmit the collected history information to the external device such as the AI server **200**. The collected history information may be used to update the learning model.

[0057] The processor **180** may control at least part of the components of AI device **100** so as to drive an application program stored in the memory **170**. Furthermore, the processor **180** may operate two or more of the components included in the AI device **100** in combination so as to drive the application program.

[0058] FIG. 2 illustrates an AI server **200** according to an embodiment of the present disclosure.

[0059] Referring to FIG. 2, the AI server **200** may refer to a device that learns an artificial neural network by using a machine learning algorithm or uses a learned artificial neural network/The AI server **200** may include a plurality of servers to perform distributed processing or may be defined as a 5G network. In this case, the AI server **200** may be included as a partial configuration of the AI device **100**, and may perform at least part of the AI processing together.

[0060] The AI server **200** may include a communication unit **210**, a memory **230**, a learning processor **240**, a processor **260**, and the like.

[0061] The communication unit **210** may transmit and receive data to and from an external device such as the AI device **100**.

[0062] The memory **230** may include a model storage unit **231**. The model storage unit **231** may store a learning or learned model (or an artificial neural network **231a**) through the learning

processor **240**.

[0063] The learning processor **240** may learn the artificial neural network **231a** by using the learning data. The learning model may be used in a state of being mounted on the AI server **200** of the artificial neural network, or may be used in a state of being mounted on an external device such as the AI device **100**.

[0064] The learning model may be implemented in hardware, software, or a combination of hardware and software. If all or part of the learning models are implemented in software, one or more instructions that constitute the learning model may be stored in memory **230**.

[0065] The processor **260** may infer the result value for new input data by using the learning model and may generate a response or a control command based on the inferred result value.

[0066] FIG. **3** illustrates an artificial intelligence apparatus applied to an energy storage system (ESS) according to one embodiment of the present disclosure.

[0067] As illustrated in FIG. **3**, the energy storage system may include a battery module **10** including a plurality of battery cells, a power storage **20**, a power generation plant **30** including solar panels, and an artificial intelligence apparatus **100** that predicts an amount of power consumption occurring in a home.

[0068] The artificial intelligence apparatus **100** may include a memory that stores power consumption data for electronic devices **40** in the home, and a processor that predicts an amount of power consumption in a home based on the power consumption data.

[0069] Here, home electronic devices **40** may include stationary electronic devices **42** such as refrigerators, washing machines, lights, and ventilators, and mobile electronic devices **44** such as electric vehicles and electric bikes.

[0070] In addition, the processor of the artificial intelligence apparatus **100** can obtain power consumption data for a pre-registered electronic device **40**, check whether there is a specific electronic device for which power consumption data has not been obtained, obtain mixed power data currently consumed in a home if there is a specific electronic device for which power consumption data has not been obtained, classifies the obtained mixed power data into a plurality of individual power data, extract individual power data matching a specific electronic device, estimate power consumption data for a specific electronic device based on the extracted individual power data, and predict an amount of power consumption in a home based on power consumption data for all pre-registered electronic devices.

[0071] Here, when the processor obtains power consumption data for a pre-registered electronic device **40**, the processor can obtain power consumption data including energy information and usage pattern information for the pre-registered electronic device.

[0072] For example, the energy information is energy information preset by the manufacturer of the electronic device **40** for the electronic device, and may include at least one of the rated voltage, power consumption, and energy consumption efficiency of the electronic device, but this is only an example and is not limited thereto.

[0073] In addition, the usage pattern information relates to the user's electronic device usage pattern of using the electronic device **40**, and may include at least one of the number of times the electronic device is used, the usage time of the electronic device, the usage cycle of the electronic device, the usage function of the electronic device, the number of times a specific function is used, the usage time of a specific function, and the usage cycle of a specific function, but this is only an example and is not limited thereto.

[0074] In some cases, the processor may obtain power consumption data corresponding to the main function of the pre-registered electronic device.

[0075] In other words, the processor can obtain the same power consumption data for electronic devices with similar main functions among pre-registered electronic devices, and can obtain different power consumption data for electronic devices with different main functions among pre-registered electronic devices.

[0076] Here, the processor can obtain power consumption data that partially overlaps in the case of electronic devices among composite electronic devices having multiple main functions, where the main functions partially overlap.

[0077] For example, a processor may obtain first power consumption data for a first electronic device performing a first main function, and obtain second power consumption data for a second electronic device performing a second main function different from the first main function, wherein energy information and usage pattern information included in the first power consumption data may be different from energy information and usage pattern information included in the second power consumption data.

[0078] For example, a processor may obtain first power consumption data for an air conditioner that performs cooling as its main function, and second power consumption data for a washing machine that performs washing as its main function, but energy information and usage pattern information included in the first power consumption data may be different from energy information and usage pattern information included in the second power consumption data.

[0079] In another case, the processor can obtain power consumption data corresponding to whether the pre-registered electronic device is capable of moving due to battery charging.

[0080] In other words, the processor can obtain the same power consumption data in the case of mobile electronic devices among the pre-registered electronic devices, and can obtain different power consumption data in the case of stationary electronic devices among the pre-registered electronic devices.

[0081] For example, the processor may obtain third power consumption data for a third mobile electronic device and obtain fourth power consumption data for a fourth stationary electronic device, and energy information and usage pattern information included in the third power consumption data may be different from energy information and usage pattern information included in the fourth power consumption data.

[0082] For example, the processor may obtain third power consumption data for an electric vehicle or an electric bike, which is a mobile electronic device **44**, and fourth power consumption data for a light or a ventilator, which is a stationary electronic device **42**, and energy information and usage pattern information included in the third power consumption data may be different from energy information and usage pattern information included in the fourth power consumption data.

[0083] In addition, when the processor obtains power consumption data for a pre-registered electronic device **40**, the processor can perform a communication connection with the pre-registered electronic device, store the power consumption data received from the electronic device connected to the communication in the memory, and obtain the power consumption data for the pre-registered electronic device from the memory.

[0084] Next, when checking whether there is a specific electronic device for which power consumption data has not been obtained, the processor can match the obtained power consumption data to a pre-registered electronic device, and check whether the power consumption data of all pre-registered electronic devices has been matched to check whether there is a specific electronic device for which power consumption data has not been obtained.

[0085] Here, the processor may recognize an electronic device among the pre-registered electronic devices **40** for which no communication connection is performed as a specific electronic device from which power consumption data is not received.

[0086] For example, in the case of electronic devices manufactured by the same manufacturer among pre-registered electronic devices **40**, communication connection is performed through the integrated app of the electronic device, so that usage pattern information of the electronic device can be obtained, but, in the case of electronic devices manufactured by other manufacturers, communication connection is not performed through the integrated app, so the manufacturer cannot obtain usage pattern information of other electronic devices.

[0087] Accordingly, the processor can recognize an electronic device among the pre-registered

electronic devices 40 for which a communication connection is not performed as a specific electronic device from which usage pattern information cannot be obtained.

[0088] Next, when the processor obtains mixed power data currently consumed within a home, the processor can obtain mixed power data in which power consumption waveforms of electronic devices consuming power at the current point in time within the home are mixed.

[0089] Next, when extracting individual power data matching a specific electronic device, the processor can input the obtained mixed power data into a pre-learned neural network model to classify it into a plurality of individual power data, and extract individual power data matching a specific electronic device among the plurality of classified individual power data.

[0090] For example, the neural network model may include an energy source classification model and an energy source estimation model.

[0091] The neural network model can input mixed power data into an energy source classification model to classify it into a number of individual power data, and input the individual power data into an energy source estimation model to estimate a specific electronic device that is the energy source of the individual power data.

[0092] Additionally, the neural network model may be pre-learned using power data of an electronic device having a specific power consumption waveform and energy information of the corresponding electronic device as input values.

[0093] Here, the neural network model may be pre-learned using energy information of electronic devices, including rated voltage, power consumption, and energy consumption efficiency, as input values.

[0094] In addition, when estimating power consumption data for a specific electronic device, the processor can estimate power consumption data including energy information and usage pattern information for the specific electronic device based on individual power data having a specific power consumption waveform.

[0095] For example, the energy information may include at least one of a rated voltage, power consumption, and energy consumption efficiency of a specific electronic device, but this is only an example and is not limited thereto.

[0096] In addition, the usage pattern information may include at least one of the number of times a specific electronic device is used, the usage time of a specific electronic device, the usage cycle of a specific electronic device, and the usage function of a specific electronic device, but this is only an example and is not limited thereto.

[0097] Here, energy information is input by a user when registering a specific electronic device and stored in memory, and usage pattern information may be estimated based on individual power data and stored in memory when extracting individual power data having a specific power consumption waveform.

[0098] In some cases, when registering a particular electronic device, the processor may obtain energy information about the particular electronic device through communication with an external server based on the product number when the user enters the product number for the particular electronic device.

[0099] Next, when predicting an amount of power consumption in a home, the processor can predict an amount of power consumption in a home by inputting first power consumption data obtained from a pre-registered electronic device and second power consumption data estimated from a specific electronic device from which the first power consumption data has not been obtained into a pre-learned neural network model.

[0100] For example, a neural network model may include a time series forecasting model that predicts the value of $t+1$ at time t .

[0101] In addition, the processor may predict an amount of power consumption in a home by additionally inputting at least one of user schedule data and external environment data into a pre-learned neural network model when predicting an amount of power consumption in a home.

[0102] Here, the processor can obtain user schedule data stored in the memory or estimate user schedule data based on user activity patterns stored in the memory and use it as an input value of the neural network model.

[0103] Additionally, the processor can obtain external environmental data including day of the week, weather, temperature, season, location, and regional information and use them as input values for the neural network model.

[0104] Next, the processor can generate prediction result information based on an amount of the predicted power consumption in a home when predicting an amount of the power consumption in a home.

[0105] Here, the processor, when generating the prediction result information, can generate the prediction result information including the usability and energy consumption status information of each electronic device in a home.

[0106] Throughout this specification, the terms neural network, network function and neural network may be used interchangeably.

[0107] The neural network model described above may be an artificial neural network (ANN) trained to output reconstructed data that is similar to the input data for the input data. An artificial neural network (ANN) is a model used in machine learning, and may refer to a model in general that has problem-solving capabilities and is composed of artificial neurons (nodes) that form a network by combining synapses.

[0108] For example, the neural network model may be an artificial neural network model based on an autoencoder. The neural network model based on an autoencoder may include, but is not limited to, an encoder part that reduces the dimensionality of data by making the number of neurons in the hidden layer smaller than the number of neurons in the input layer, and a decoder part that reconstructs the data by expanding the dimensionality of data from the hidden layer again, and has an output layer having the same number of neurons as the number of neurons in the input layer.

[0109] In addition, the neural network model may be an artificial neural network model based on a generative adversarial network (GAN). A generative adversarial network (GAN) may be an artificial neural network in which a generator and a discriminator are learned adversarially, but is not limited thereto.

[0110] In addition, the neural network model may be a deep neural network.

[0111] A deep neural network (DNN) may refer to a neural network that includes a plurality of hidden layers in addition to an input layer and an output layer. Using a deep neural network, latent structures of data can be identified. In other words, latent structures of photos, text, videos, voices, and music (for example, what objects are in the photo, what the content and emotion of the text are, what the content and emotion of the voice are, or the like) can be identified. A deep neural network may include a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a Q network, a U network, a Siamese network, or the like.

[0112] In this way, the artificial intelligence apparatus of the present disclosure can accurately predict an amount of power consumption in a home by estimating power consumption data of an electronic device whose usage pattern is unknown using a neural network model.

[0113] In addition, the artificial intelligence apparatus of the present disclosure can provide accurate energy consumption information for each electronic device to the user by predicting an amount of power consumption in a home based on energy information and usage pattern information for the electronic device.

[0114] In addition, the artificial intelligence apparatus of the present disclosure can provide accurate, specific, and highly reliable energy consumption information in a home to the user by predicting an amount of power consumption in a home based on power consumption data of each electronic device, user schedule data, and external environment data using a neural network model.

[0115] FIG. 4 is a view for explaining an electronic device registration process for predicting an

amount of power consumption according to one embodiment of the present disclosure.

[0116] As illustrated in FIG. 4, the artificial intelligence apparatus **100** can receive a registration request for electronic devices for predicting an amount of power consumption from a user terminal **50**.

[0117] Here, the user terminal **50** can display an electronic device registration window **60** on the display screen through an electronic device integrated app, register a number of electronic devices for predicting an amount of power consumption through the electronic device registration window **60**, and then provide information on the registered electronic devices to the artificial intelligence apparatus **100**.

[0118] In some cases, the artificial intelligence apparatus **100** may also receive user input for a request to register an electronic device through a user interface.

[0119] In another case, the artificial intelligence apparatus **100** may obtain product information from electronic devices capable of communication within the home, obtain corresponding energy information from an external server, and then automatically provide the product information and energy information of the obtained electronic devices to a registration window **60** displayed on the user terminal **50**.

[0120] In this case, the user may simply and quickly input only product information about the electronic device that is not obtained from the artificial intelligence apparatus **100** through the registration window **60** displayed on the user terminal **50**.

[0121] For example, the artificial intelligence apparatus **100** can obtain product information from air conditioners, washing machines, and air purifiers that are capable of communicating within a home and are manufactured by the same manufacturer, obtain corresponding energy information from the server of the manufacturer, and then automatically provide the product information and energy information of the obtained electronic devices to the registration window **60** displayed on the user terminal **50**.

[0122] In addition, the user can register the vacuum cleaner by entering only product information **62** about another manufacturer's vacuum cleaner that is not obtained by the artificial intelligence apparatus **100** through the registration window **60** displayed on the user terminal **50**.

[0123] Next, the user terminal **50** can register product information about the vacuum cleaner by transmitting it to the artificial intelligence apparatus **100** through user input by touching the registration button **64** of the registration window **60**.

[0124] Here, the artificial intelligence apparatus **100** can estimate power consumption data such as usage patterns by additionally obtaining energy information of the vacuum cleaner corresponding to the product model number by communicating with an external server using only the product model number among the product information on the vacuum cleaner.

[0125] The artificial intelligence apparatus **100** can obtain power consumption data including energy information and usage pattern information for pre-registered electronic devices.

[0126] For example, the energy information may include at least one of the rated voltage, power consumption, and energy consumption efficiency of electronic devices, including an air conditioner **40a**, a washing machine **40b**, an air purifier **40c**, and a vacuum cleaner **40d**.

[0127] In addition, the usage pattern information may include at least one of the number of times the electronic device is used, the usage time of the electronic device, the usage cycle of the electronic device, the usage function of the electronic device, the number of times a specific function is used, the usage time of a specific function, and the usage cycle of a specific function.

[0128] In some cases, the artificial intelligence apparatus **100** may obtain power consumption data corresponding to the main function of a pre-registered electronic device.

[0129] In other words, the artificial intelligence apparatus **100** can obtain the same power consumption data for electronic devices with similar main functions among pre-registered electronic devices, and can obtain different power consumption data for electronic devices with different main functions among pre-registered electronic devices.

[0130] Here, the artificial intelligence apparatus **100** can obtain power consumption data that partially overlaps in the case of electronic devices among composite electronic devices having multiple main functions, in which some of the main functions overlap.

[0131] In another case, the artificial intelligence apparatus **100** can obtain power consumption data corresponding to whether movement is possible due to battery charging of a pre-registered electronic device.

[0132] In other words, the artificial intelligence apparatus **100** can obtain the same power consumption data in the case of mobile electronic devices among pre-registered electronic devices, and can obtain different power consumption data in the case of stationary electronic devices among pre-registered electronic devices.

[0133] In addition, the artificial intelligence apparatus **100** can perform a communication connection with a pre-registered electronic device and obtain power consumption data received from the electronic device to which the communication is connected.

[0134] Next, the artificial intelligence apparatus **100** matches the obtained power consumption data to pre-registered electronic devices and checks whether the power consumption data is matched to all pre-registered electronic devices to check whether there is a specific electronic device for which power consumption data has not been obtained.

[0135] Here, the artificial intelligence apparatus **100** can recognize an electronic device among pre-registered electronic devices in which a communication connection is not performed as a specific electronic device from which power consumption data is not received.

[0136] FIGS. **5** and **6** are views for explaining an energy source estimation process of a neural network model according to an embodiment of the present disclosure.

[0137] As illustrated in FIGS. **5** and **6**, the artificial intelligence apparatus **100** may obtain mixed power data consumed by a plurality of electronic devices in the current household if there is a specific electronic device for which power consumption data has not been obtained, input the obtained mixed power data into a pre-learned neural network model **70** to classify it into a plurality of individual power data, and estimate individual power data consumed by the specific electronic device based on the classified individual power data.

[0138] Here, the artificial intelligence apparatus **100** can obtain mixed power data in which the power consumption waveforms of electronic devices consuming power at the current point in time in the home are mixed.

[0139] In addition, the artificial intelligence apparatus **100** can input the obtained mixed power data into a pre-learned neural network model **70** to classify it into a plurality of individual power data, and extract individual power data matching a specific electronic device from among the classified plurality of individual power data.

[0140] As illustrated in FIG. **5**, mixed power data in which power consumption waveforms of an air conditioner **40a**, a washing machine **40b**, an air purifier **40c**, and a vacuum cleaner **40d** among electronic devices are mixed is input into a pre-learned neural network model **70** to be classified into a plurality of individual power data, and based on the classified individual power data, first individual power data consumed by the air conditioner **40a**, second individual power data consumed by the washing machine **40b**, third individual power data consumed by the air purifier **40c**, and fourth individual power data consumed by the vacuum cleaner **40d** can be extracted.

[0141] Accordingly, the artificial intelligence apparatus **100** can estimate the usage pattern of the vacuum cleaner **40d** based on the fourth individual power data consumed by the vacuum cleaner **40d** when the specific electronic device is a vacuum cleaner.

[0142] As illustrated in FIG. **6**, the neural network model **70** may include an energy source classification model **72** and an energy source estimation model **74**.

[0143] The artificial intelligence apparatus **100** can input mixed power data into an energy source classification model **72** to classify it into a plurality of individual power data, and input the individual power data into an energy source estimation model **74** to estimate a specific electronic

device that is the energy source of the individual power data.

[0144] Additionally, the neural network model **70** can be pre-learned using power data of an electronic device having a specific power consumption waveform and energy information of the corresponding electronic device as input values.

[0145] Here, the neural network model **70** can be pre-learned using energy information of the electronic device, including rated voltage, power consumption, and energy consumption efficiency, as input values.

[0146] In addition, the artificial intelligence apparatus **100** can estimate power consumption data including energy information and usage pattern information for a specific electronic device based on individual power data having a specific power consumption waveform.

[0147] For example, the energy information may include at least one of the rated voltage, power consumption, and energy consumption efficiency of a specific electronic device.

[0148] Additionally, the usage pattern information may include at least one of the number of times a specific electronic device is used, the usage time of a specific electronic device, the usage cycle of a specific electronic device, and the usage function of a specific electronic device.

[0149] Here, energy information is input by a user when registering a specific electronic device and stored in memory, and usage pattern information can be estimated based on individual power data and stored in memory when extracting individual power data having a specific power consumption waveform.

[0150] In some cases, when registering a specific electronic device, the artificial intelligence apparatus **100** may obtain energy information about a specific electronic device through communication with an external server based on the product number when the user enters the product number for the specific electronic device.

[0151] FIG. **7** is a view for explaining a process of predicting an amount of power consumption in a home using a neural network model according to an embodiment of the present disclosure.

[0152] As illustrated in FIG. **7**, the artificial intelligence apparatus **100** can predict an amount of power consumption in a home based on power consumption data for all pre-registered electronic devices.

[0153] Here, the artificial intelligence apparatus **100** can predict an amount of power consumption in a home by inputting first power consumption data obtained from a pre-registered electronic device and second power consumption data estimated from a specific electronic device from which the first power consumption data has not been obtained into a pre-learned neural network model.

[0154] For example, a neural network model may include a time series forecasting model that predicts the value of $t+1$ at time t .

[0155] In addition, the artificial intelligence apparatus **100** can predict an amount of power consumption in a home by additionally inputting at least one of user schedule data and external environment data into a pre-learned neural network model.

[0156] Here, the artificial intelligence apparatus **100** can obtain user schedule data stored in memory or estimate user schedule data based on user activity patterns stored in memory and use it as an input value of a neural network model.

[0157] Additionally, the artificial intelligence apparatus **100** can obtain external environmental data including day of the week, weather, temperature, season, location, and regional information and use them as input values for a neural network model.

[0158] FIG. **8** is a view for explaining information on prediction results of an amount of power consumption in a home according to one embodiment of the present disclosure.

[0159] As illustrated in FIG. **8**, the artificial intelligence apparatus **100** can generate prediction result information based on an amount of the predicted power consumption within the home.

[0160] Here, the artificial intelligence apparatus **100** can generate prediction result information including information on the usability and energy consumption status of each electronic device in a home.

[0161] For example, the artificial intelligence apparatus **100** can generate prediction result information and provide a prediction result of an amount of power consumption in a home to a user terminal **50**.

[0162] The user terminal **50** can provide prediction result information including information on the usability and energy consumption status of each electronic device in a home through a prediction information window **80** of an amount of power consumption displayed on the display screen.

[0163] Here, the prediction information window **80** of an amount of power consumption can provide prediction result information of an amount of power consumption by period based on user schedule data, external environment data, and power consumption data for each electronic device, and can additionally provide a notification message for an electronic device for which an increase in an amount of power consumption is predicted.

[0164] FIG. **9** is a flowchart for explaining a method for predicting power consumption of an artificial intelligence apparatus according to an embodiment of the present disclosure.

[0165] As illustrated in FIG. **9**, the artificial intelligence apparatus can obtain power consumption data for a pre-registered electronic device (**S10**).

[0166] Here, the artificial intelligence apparatus can obtain power consumption data including energy information and usage pattern information for pre-registered electronic devices.

[0167] In some cases, the artificial intelligence apparatus may obtain power consumption data corresponding to the main function of a pre-registered electronic device.

[0168] In another case, the artificial intelligence apparatus may obtain power consumption data corresponding to whether the pre-registered electronic device is capable of moving due to battery charging.

[0169] In addition, the artificial intelligence apparatus may perform a communication connection with a pre-registered electronic device, store power consumption data received from the electronic device to which the communication is connected in a memory, and obtain power consumption data for the pre-registered electronic device from the memory.

[0170] Next, the artificial intelligence apparatus can check whether there is a specific electronic device for which power consumption data has not been obtained (**S20**).

[0171] Here, the artificial intelligence apparatus can match the obtained power consumption data to pre-registered electronic devices and check whether the power consumption data of all pre-registered electronic devices are matched to check whether there is a specific electronic device for which power consumption data has not been obtained.

[0172] The artificial intelligence apparatus can recognize a specific electronic device among pre-registered electronic devices that is not performing a communication connection as one from which power consumption data is not received.

[0173] In addition, the artificial intelligence apparatus can obtain mixed power data currently consumed in the home if there is a specific electronic device for which power consumption data has not been obtained (**S30**).

[0174] Here, the artificial intelligence apparatus can obtain mixed power data in which power consumption waveforms of electronic devices consuming power at the current point in time in a home are mixed.

[0175] Next, the artificial intelligence apparatus can classify the obtained mixed power data into a plurality of individual power data and extract individual power data matching a specific electronic device (**S40**).

[0176] Here, the artificial intelligence apparatus can input the obtained mixed power data into a pre-learned neural network model to classify it into a plurality of individual power data, and extract individual power data matching a specific electronic device from among the classified plurality of individual power data.

[0177] Next, the artificial intelligence apparatus can estimate power consumption data for a specific electronic device based on the extracted individual power data (**S50**).

[0178] Here, the artificial intelligence apparatus can estimate power consumption data including energy information and usage pattern information for a specific electronic device based on individual power data having a specific power consumption waveform.

[0179] Additionally, the artificial intelligence apparatus can predict an amount of power consumption in a home based on power consumption data for all pre-registered electronic devices (S60).

[0180] Here, the artificial intelligence apparatus can predict an amount of power consumption in a home by inputting first power consumption data obtained from a pre-registered electronic device and second power consumption data estimated from a specific electronic device from which the first power consumption data has not been obtained into a pre-learned neural network model.

[0181] In some cases, the artificial intelligence apparatus may also predict an amount of power consumption in a home by additionally inputting at least one of user schedule data and external environment data into a pre-trained neural network model.

[0182] Here, the artificial intelligence apparatus can obtain user schedule data stored in memory or estimate user schedule data based on user activity patterns stored in memory and use the data as input values for the neural network model, and can obtain external environment data including day of the week, weather, temperature, season, location, and regional information and use the data as input values for the neural network model.

[0183] Additionally, the artificial intelligence apparatus can generate prediction result information based on an amount of the predicted power consumption in a home.

[0184] Here, the prediction result information may include information on the usability and energy consumption status of each electronic device in the home.

[0185] In this way, the artificial intelligence apparatus of the present disclosure can accurately predict power consumption in a home by estimating power consumption data of electronic devices whose usage patterns are unknown using a neural network model.

[0186] In addition, the artificial intelligence apparatus of the present disclosure can provide accurate energy consumption information for each electronic device to the user by predicting power consumption in a home based on energy information and usage pattern information for the electronic device.

[0187] In addition, the artificial intelligence apparatus of the present disclosure can provide accurate, specific, and highly reliable home energy consumption information to the user by predicting an amount of power consumption in a home based on power consumption data of each electronic device, user schedule data, and external environment data using a neural network model.

INDUSTRIAL APPLICABILITY

[0188] The artificial intelligence apparatus according to the present disclosure has remarkable industrial applicability because it has the effect of accurately predicting an amount of power consumption in a home by estimating power consumption data of an electronic device whose usage pattern is unknown using a neural network model.

Claims

1. An artificial intelligence apparatus comprising: a memory storing power consumption data for electronic devices in a home; and a processor predicting power consumption in a home based on the power consumption data, wherein the processor obtains power consumption data for pre-registered electronic devices, checks whether there is a specific electronic device for which the power consumption data has not been obtained, and if there is a specific electronic device for which the power consumption data has not been obtained, obtains mixed power data consumed in a current home, classifies the obtained mixed power data into a plurality of individual power data, extracts individual power data matching the specific electronic device, estimates power consumption data for the specific electronic device based on the extracted individual power data,

and predicts an amount of power consumption in a home based on power consumption data for all pre-registered electronic devices.

2. The artificial intelligence apparatus of claim 1, wherein the processor, when obtaining power consumption data for the pre-registered electronic device, obtains power consumption data including energy information and usage pattern information for the pre-registered electronic device.
3. The artificial intelligence apparatus of claim 1, wherein the processor, when obtaining power consumption data for the pre-registered electronic device, performs a communication connection with the pre-registered electronic device, stores the power consumption data received from the electronic device connected to the communication in the memory, and obtains the power consumption data for the pre-registered electronic device from the memory.
4. The artificial intelligence apparatus of claim 1, wherein the processor, when checking whether there is a specific electronic device for which the power consumption data has not been obtained, matches the obtained power consumption data to the pre-registered electronic device, and checks whether the power consumption data has been matched to all of the pre-registered electronic devices to check whether there is a specific electronic device for which the power consumption data has not been obtained.
5. The artificial intelligence apparatus of claim 4, wherein the processor recognizes an electronic device among the pre-registered electronic devices in which a communication connection is not performed as a specific electronic device from which the power consumption data is not received.
6. The artificial intelligence apparatus of claim 1, wherein the processor, when obtaining mixed power data consumed in the current home, obtains mixed power data in which power consumption waveforms of electronic devices consuming power at the current point in time in the home are mixed.
7. The artificial intelligence apparatus of claim 1, wherein the processor, when extracting individual power data matching the specific electronic device, inputs the obtained mixed power data into a pre-learned neural network model to classify the obtained mixed power data into a plurality of individual power data, and extracts individual power data matching the specific electronic device from among the classified plurality of individual power data.
8. The artificial intelligence apparatus of claim 7, wherein the neural network model is pre-learned with power data of an electronic device having a specific power consumption waveform and energy information of the corresponding electronic device as input values.
9. The artificial intelligence apparatus of claim 7, wherein the neural network model is pre-learned with energy information of an electronic device including rated voltage, power consumption, and energy consumption efficiency as input values.
10. The artificial intelligence apparatus of claim 1, wherein the processor, when estimating power consumption data for the specific electronic device, estimates power consumption data including energy information and usage pattern information for the specific electronic device based on individual power data having a specific power consumption waveform.
11. The artificial intelligence apparatus of claim 1, wherein the processor, when predicting the amount of power consumption in a home, inputs the first power consumption data obtained from a pre-registered electronic device and the second power consumption data estimated from a specific electronic device from which the first power consumption data has not been obtained into a pre-learned neural network model to predict the amount of power consumption in a home.
12. The artificial intelligence apparatus of claim 11, wherein the neural network model includes a time series forecasting model that predicts the $t+1$ value at time t .
13. The artificial intelligence apparatus of claim 11, wherein the processor, when predicting the amount of power consumption in a home, at least one of the user schedule data and the external environment data is additionally input to a pre-learned neural network model to predict the amount of power consumption in a home.
14. The artificial intelligence apparatus of claim 1, wherein the processor, when predicting the

amount of power consumption in a home, generates prediction result information based on an amount of the predicted power consumption in a home.

15. A method for predicting power consumption of an artificial intelligence apparatus including a processor for predicting an amount of power consumption in a home, comprising: obtaining power consumption data for a pre-registered electronic device; checking whether there is a specific electronic device for which the power consumption data has not been obtained; obtaining mixed power data consumed in a current home if there is a specific electronic device for which the power consumption data has not been obtained; classifying the obtained mixed power data into a plurality of individual power data and extracting individual power data matching the specific electronic device; estimating power consumption data for the specific electronic device based on the extracted individual power data; and predicting the amount of power consumption in a home based on power consumption data for all the pre-registered electronic devices.
